

Project Summary (Developer Perspective)

The Real Estate Facility Management System is a web-based application designed to replace manual and fragmented property management processes with a centralized, automated platform. It targets property managers, administrators, and tenants by integrating core operations such as property management, tenant handling, rent tracking, and maintenance coordination into a single system.

The system follows a client-server architecture and is implemented using a typical web stack (HTML, CSS, JavaScript, PHP, MySQL). It adopts a modular design and follows the Waterfall development model, ensuring structured development from requirements to deployment.

Aim of the Project

The main aim is to design and implement a secure, scalable, and user-friendly web-based system that automates real estate and facility management operations, improves efficiency, and supports better decision-making through centralized data management.

Core Objectives

From a development standpoint, the system is expected to:

- Build an integrated platform for managing properties, tenants, and facility data
- Implement a rent management system to track payments, balances, and history
- Develop a maintenance module for request submission, tracking, and resolution
- Provide a user-friendly interface for different roles (admin, manager, tenant)
- Enforce authentication and role-based access control for security
- Ensure data integrity and consistency using relational database design
- Evaluate system performance, usability, and reliability after implementation

System Requirements

Functional Requirements

User authentication (login/register with roles: admin, manager, tenant)

- Property management (add, update, track availability/status)
- Tenant management (assign tenants to properties, manage records)
- Rent management (record payments, track balances, generate history)
- Maintenance management (submit, track, update requests)
- Reporting (generate system-wide reports on operations)

Non-Functional Requirements

- Security: authentication, authorization, and data protection
- Performance: fast response time and efficient data processing
- Usability: simple, intuitive UI/UX
- Reliability: consistent operation with minimal downtime

Technical Stack / Implementation Requirements

Hardware Requirements

- CPU: Dual-core 2.0 GHz or higher
- RAM: Minimum 4 GB
- Storage: 500 GB HDD/SSD
- Network: Stable internet connection
- Client device: Desktop or laptop

Software Requirements

- OS: Windows / Linux / macOS
- Frontend: HTML, CSS, JavaScript
- Backend: PHP
- Database: MySQL
- Web Server: Apache
- Development Tool: VS Code
- Browser: Chrome / Firefox / Edge

System Architecture

- **Presentation Layer:** UI (HTML, CSS, JS)
- **Application Layer:** Business logic (PHP)
- **Data Layer:** Database (MySQL)

Modules

- User Management (authentication & roles)
- Property Management
- Tenant Management
- Rent Management
- Maintenance Management
- Reporting & Admin Control